COMPUTATIONAL STRUCTURAL ANALYSIS VERIFICATION REPORT

Project Code: AS-2026-TRUSS

Assessment Target: Roof Support Truss Cross-Sections

Compliance Standard: AISC Steel Construction Manual Specification

1. Objective

This engineering document compiles structural element evaluation checks performed on critical node connections. Static axial loading configurations were parsed using fixed finite matrix stress transformations to guarantee compliance with structural safety standards.

2. Theoretical Mechanics Framework

Axial member integrity checking uses deterministic structural evaluations:

-Normal Axial Stress (σ): Computed directly via $\sigma = \frac{F}{A}$, where F indicates element axial force and A describes gross profile cross-sectional geometric layout values.

-Structural Safety Margin (M):Defined by the expression :

$$M = \frac{\sigma_{allowable}}{\sigma_{calculated}} - 1$$

Negative evaluations show stress demand exceeds structural capacity bounds, marking target members for immediate design reassessment.

3. Structural Evaluation Findings

Computational runs revealed localized stress concentration exceptions:

Member 101: Met baseline standard requirements, exhibiting structural compliance margins within expected parameters.

-Member 102: Exceeded permissible code boundaries under extreme force configurations, triggering a structural failure flag.

Member 103: Confirmed structural stability under compressive stress profiles.

4. Remedial Design Directives

Member 102 Redesign: Increase member profile thickness to elevate cross-sectional layout area A to at least 0.0025 square meters.

Material Update:Substitute standard A36 profiles with high-strength structural grade steels where geometry modification is restricted by architectural clearances.

CODE OUTPUT:

```
   Elem ID      Stress (MPa)         Margin                   Status
------------------------------------------------------------------
       101             80000         -0.997  STRUCTURAL_FAILURE_RISK
       102          2.92e+05         -0.999  STRUCTURAL_FAILURE_RISK
       103            -3e+04         -0.992  STRUCTURAL_FAILURE_RISK

Process returned 0 (0x0)   execution time : 0.332 s
Press any key to continue.
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